Linux Network Emulation and SDN Lab Using Mininet

Overview

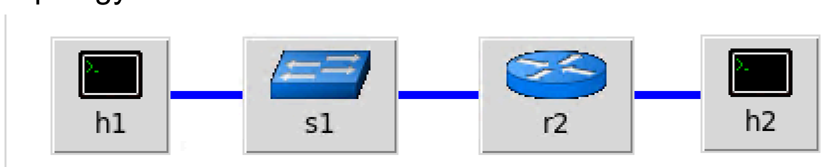
Designed and tested a virtual software-defined networking (SDN) environment using the Mininet VM and MiniEdit. Configured hosts, switches, and router interfaces, analyzed routing behavior, tested inter-network communication, monitored TCP/UDP sockets, and implemented Linux firewall rules using UFW.

Environment & Tools

- Platform: Mininet VM
- Network Emulator: Mininet
- Topology Editor: MiniEdit
- Operating System: Linux
- Networking Utilities: ifconfig, route, ping, ss, ps
- Traffic Testing Tool: iPerf
- Firewall Management: UFW (Uncomplicated Firewall)

Network Topology

Topology created and tested:



Device Roles

- h1 / h2: End hosts
- s1: Layer 2 switch
- r2: Router connecting two subnets

Technical Tasks Performed

1. Interface and MAC Address Analysis

Identified router interfaces and hardware addresses using Linux networking commands.

Router Interfaces	
Interface	MAC Address

r2-eth0	d6:c2:2a:70:cc:7f
r2-eth1	32:26:63:6b:41:16

Skills Demonstrated

- Linux network interface analysis
- MAC address identification
- CLI-based troubleshooting

2. Routing Table Configuration & Analysis

Examined routing behavior and interpreted CIDR-based routing entries.

Router Table		
Destination Network	CIDR	Outgoing Interface
10.0.0.0	/8	r2-eth0
11.0.0.0	/8	r2-eth1

Concepts Applied

- Static routing
- CIDR notation
- Subnet analysis
- Route interpretation

3. Network Connectivity Testing

Used ICMP ping testing to verify communication between hosts and router interfaces.

Findings

- Initial communication between `h2` and `11.0.0.10` failed due to missing routing configuration.

- After reconfiguring `h2` into the `11.0.0.0/8` subnet, communication succeeded.
- Verified successful routing between separate subnets through `r2`.

Skills Demonstrated

- Network troubleshooting
- Layer 3 routing analysis
- Gateway configuration
- ICMP diagnostics

4. Default Gateway & Inter-Subnet Routing

Configured default gateways to enable communication between separate networks.

Commands Used

- route add default gw 10.0.0.10
- route add default gw 11.0.0.10

Outcome

Successfully enabled communication between:

- `10.0.0.0/8`
- `11.0.0.0/8`

Concepts Demonstrated

- Default gateway configuration
- Router-based packet forwarding
- Inter-network communication

5. Switch Behavior Analysis

Verified Layer 2 switch functionality within the topology.

Key Observation

Confirmed that the switch `s1` could not be pinged directly because:

- Layer 2 switches do not use IP addresses by default
- Ping requires Layer 3 IP communication

Networking Concepts

- OSI model understanding
- Layer 2 vs Layer 3 communication
- Ethernet switching fundamentals

6. TCP & UDP Socket Analysis

Analyzed active network sockets using the Linux `ss` utility.

TCP Socket States Observed

- LISTEN
- ESTABLISHED
- TIME-WAIT

UDP Socket States Observed

- UNCONN

Skills Demonstrated

- Socket inspection
- TCP state analysis
- UDP behavior analysis
- Linux network diagnostics

7. Process & Port Investigation

Mapped active sockets to running processes using Linux process management tools.

Identified Services

Protocol	Port	Process	PID
UDP	5001	iperf	3140
TCP	5001	iperf	3137

Commands Used

- ss -a
- ps u -p <PID>

Skills Demonstrated

- Linux process analysis
- Service identification
- Port-to-process mapping

8. Firewall Configuration & Testing

Configured Linux firewall rules using UFW and verified access control behavior.

Firewall Actions

- Enabled firewall protection: ufw enable
- Allowed iPerf TCP traffic: ufw allow 5001/tcp

Observations

- Initial iPerf connection attempts failed due to blocked incoming traffic.
- After adding firewall rules, TCP connectivity succeeded.

Skills Demonstrated

- Linux firewall management
- Access control configuration
- Network security testing
- Service rule implementation

Key Networking Concepts Applied

- TCP/IP Networking
- CIDR Subnetting
- Static Routing
- Default Gateways
- Layer 2 Switching
- Layer 3 Routing
- TCP/UDP Socket States
- Linux Firewall Administration
- Network Troubleshooting
- Client-Server Communication

Appendix / Technical Evidence

Appendix A: Routing Table Verification

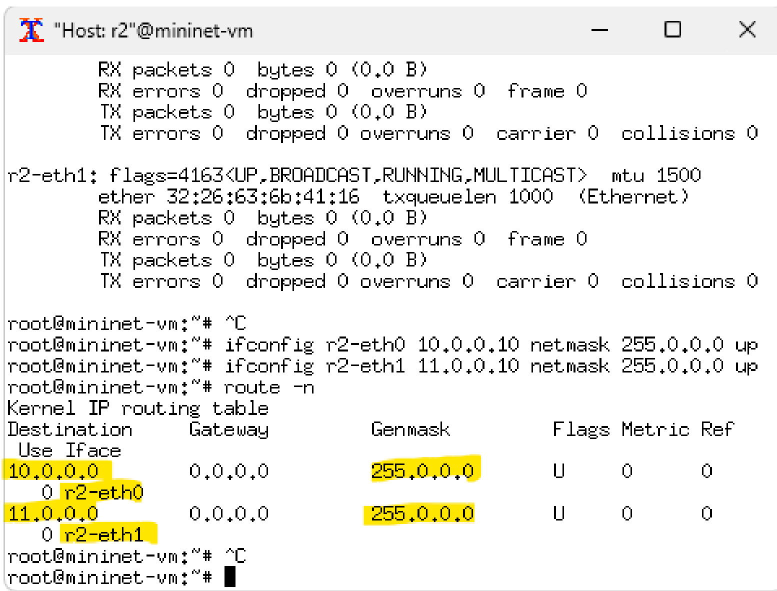


Figure 1: Routing table on `r2` showing routes for `10.0.0.0/8` and `11.0.0.0/8`

Appendix B: Successful Inter-Subnet Ping

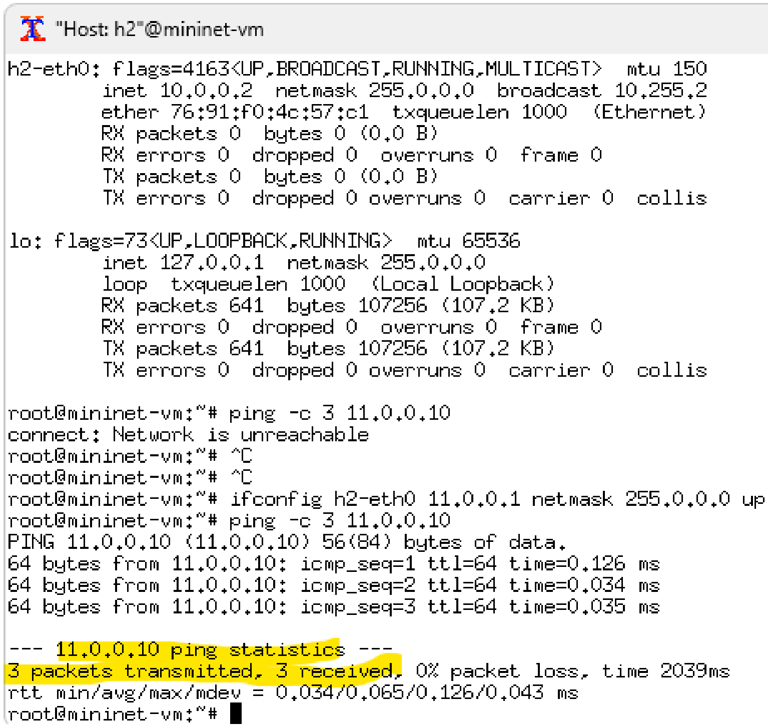


Figure 2: ICMP communication between hosts after default gateway configuration

Appendix C: Firewall Rule Configuration

Figure 3: UFW firewall rule allowing TCP traffic on port 5001